

siam_vqe — Progress and Results Report
VQE for the NiO single-impurity Anderson model, validated against EDRIXS

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1. Overview and status

`siam_vqe` is a Qiskit variational-quantum-eigensolver (VQE) study of the NiO single-impurity Anderson model (SIAM), parameterised from Haverkort *et al.*, *Phys. Rev. B* **85**, 165113 (2012) / EDRIXS pedagogical examples 3 and 6, and validated end-to-end against exact diagonalisation (ED). The project climbs an **active-space ladder** **L0** $\rightarrow$ **L3**, adds **spectroscopy** (quantum equation-of-motion, qEOM, and L-edge XAS), and culminates in a **real-hardware execution track** on IBM Quantum (`ibm_marrakesh`).

Every phase is gated by a fixed **validation-layer framework** (Section 3) and implemented test-first; the full suite stands at **256 passed / 1 xfailed**.

1.1 Phase status

Phase	Scope	Active space	Qubits	Result	Status
1 — L0	Hubbard dimer	2 sites $\times$ 2 spin	2 (tapered)	$\Delta E = 5.7 \times 10^{-9}$	merged
2 — L1	impurity-d + ligand-p	4 spin-orbitals	2 (tapered)	noiseless $\Delta E = 1.1 \times 10^{-12}$; on hardware	merged
3 — L2	e_g + bath, noise study	6	6	M3+ZNE recovers 92% of the gap	merged
4 — L3	full Ni 3d + ligand bath, ADAPT	20 spin-orbitals	18	$ \Delta E = 9.1 \times 10^{-11}$	merged
5 — qEOM + XAS	excited states, L-edge XAS	—	18	L_2 vs ED $\approx 4 \times 10^{-6}$	merged
5b — Hardware	ISA-correct M3/ZNE, real run	4	5 (+anc)	4/4 ISA-dispatch; noise-dominated	merged

Repository `liamlts/research-cowork`, package `siam_vqe` (v0.4.0); `main` at merge commit 648c125c after PR #4 (Phase 5b) and PR #5 (`_retranslate_to_basis` Qiskit-2.0 cleanup).

2. System and parameters

The model is an Anderson impurity: a correlated Ni 3d shell hybridised with a ligand bath, with on-site Coulomb (Slater–Condon / Kanamori) interactions and a crystal field. The L3 (full) Hamiltonian uses EDRIXS example-3 parameters verbatim:

Parameter	Value (eV)	Meaning
F_{dd}^2	9.787	Slater integral (d–d)
F_{dd}^4	6.078	Slater integral (d–d)

Parameter	Value (eV)	Meaning
U_{dd}	7.3	monopole Coulomb
Δ	4.7	charge-transfer energy
V_{e_g}	2.06	e_g hybridisation
$V_{t_{2g}}$	1.21	t_{2g} hybridisation
$10Dq$	0.56	cubic crystal field

Ni $d^8 + 10$ bath electrons = 18 electrons in 20 spin-orbitals; the (9, 9) particle sector is tapered to 18 qubits. The reduced levels L0–L2 use consistent sub-spaces (Section 4).

3. Methods and the validation-layer framework

- **Mapping / tapering.** Fermionic operators are Jordan–Wigner mapped; symmetry (parity / particle-number) tapering reduces qubit count (e.g. L1: 4 spin-orbitals $\rightarrow$ 2 qubits).
- **Ansätze.** Fixed UCCSD or **EfficientSU2** at the small levels; **ADAPT-VQE** at L3 (a 99-operator generalised-singles/doubles pool, gradient-screened, threshold 10^{-4} eV).
- **Reference.** A `scipy.sparse` ED in the same sector is the ground truth (“Path B”); EDRIXS provides an independent cross-check (“Path A”).
- **Error mitigation** (hardware path). Manual M3 readout correction (mthree) + digital Zero-Noise Extrapolation (ZNE) by global circuit folding.

Each phase must pass a numbered ladder of checks (pass-or-flag where noted):

1. **Energy match** — $|E_{VQE} - E_{ED}| < \text{tol}$.
2. **State overlap** — $|\langle\psi_{ED}|\psi_{VQE}\rangle|^2 \rightarrow 1$ (soft-degeneracy mode if the ED ground state is near-degenerate).
3. **Expressivity** — the ansatz manifold contains the target.
4. **Multistart spread** — convergence across seeds.
5. **Observable agreement** — $\langle n_d \rangle, \langle n_p \rangle, \langle S^2 \rangle$, per-orbital occupations.
6. **Resilience guardrail** (hardware, pass-or-flag) — mitigation should not worsen the ED gap.
7. **Mitigation effectiveness** (L2, hard gate) — best config’s gap ratio < 1 .

4. Phase L0 — Hubbard dimer (foundation)

A two-site Hubbard dimer ($U = 4$, $t = 1$, half-filling) on 2 tapered qubits, **EfficientSU2** (reps 3, 16 parameters), COBYLA. Establishes the package, the tapering, and the validation harness.

- $E_{VQE} = -0.82842712$, $E_{ED} = -0.82842712 \rightarrow \Delta E = 5.7 \times 10^{-9}$; overlap = 0.99999999. All five layers pass at near-machine precision.

5. Phase L1 — minimal NiO + first hardware

L1 keeps one impurity-d and one ligand-p orbital (4 spin-orbitals $\rightarrow$ 2 tapered qubits), $U = 7.3$, $V = 2.06$, with EDRIXS charge-transfer levels ($\varepsilon_d = -40.85$, $\varepsilon_p = +13.38$).

- **Noiseless:** $E_{VQE} = -74.58626153$ vs $E_{ED} = -74.58626153 \rightarrow \Delta E = 1.1 \times 10^{-12}$; multistart spread 1.4×10^{-8} . UCCSD needs only 3 tiny parameters (≈ 0.04 rad); the circuit is depth 14

with **2 CZ gates**.

- **Noisy simulator** (`AerSimulator.from_backend(FakeMarrakesh)`): $E = -73.97$ ($\sigma = 0.15$).
- **Real ibm_marrakesh** (2026-05-25): un-mitigated $E = -72.40$ ($\sigma = 0.12$), M3 $E = -74.04$ ($\sigma = 0.13$), M3+ZNE $E = -73.96$ ($\sigma = 0.35$).
- **Layer 6 (resilience guardrail): FAILED, honestly reported.** M3 cut the ED gap $\sim 4\times$ ($2.19 \rightarrow 0.55$ eV) but M3+ZNE made it slightly *worse* ($0.55 \rightarrow 0.62$ eV) with a $3\times$ wider error bar. ZNE on a 2-CZ near-identity circuit is variance-dominated.

Physics note. With the EDRIXS charge-transfer levels the (1,1)-sector ground state is $d^2\bar{L}^0$ -dominated ($\langle n_d \rangle = 1.996$, $\langle n_p \rangle = 0.004$) — essentially both electrons on the impurity, hybridisation $< 1\%$. The hardware run therefore measured device noise on a near-identity circuit, not Kondo physics; a particle-hole-symmetric gauge would be needed for a “real Kondo on hardware” demo.

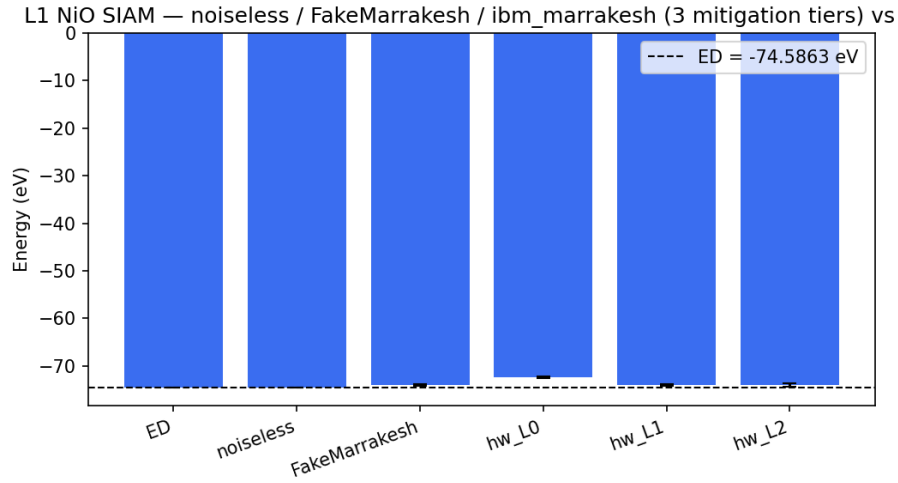


Figure 1: L1 energy comparison: noiseless, noisy-sim, and three hardware mitigation tiers against the ED reference.

6. Phase L2 — noise study and mitigation grid

L2 opens the $e_g + \text{bath}$ active space (6 qubits) and is the dedicated **error- mitigation study** on a noisy simulator (8192 shots). The Kanamori interaction uses the **STK Racah convention** ($U = 8.595$, $J_H = 1.006$, $U' = 6.584$ eV), chosen to pin the $^1A_{1g}$ multiplet gap exactly. ED = -95.400 ; noiseless VQE = -95.400 .

An 8-configuration mitigation grid (no-mit, M3, ZNE, and M3+ZNE with linear / exponential / polynomial-degree-3 extrapolators over several noise-factor schedules) was swept.

- **Layer 6 (passed):** M3+ZNE gap 2.94 eV $\ll$ M3-only gap 14.33 eV.
- **Layer 7 (hard gate, passed):** best config `m3_zne_poly3_135`, gap ratio **0.0755** ($\approx 92\%$ noise reduction).

Config (gap ratio, lower = better)	ratio
m3_zne_poly3_135	0.0755

Config (gap ratio, lower = better)	ratio
m3_zne_exp_135	0.1947
m3_zne_lin_123	0.3373
m3_zne_lin_12345	0.5904
m3_zne_lin_135	0.6339
zne_lin_135	0.6461
m3_only	0.9500

Finding: nonlinear extrapolators dominate linear ones; the exponential fit is under-determined on 3 noise points (poly-3 is the more reliable headline).

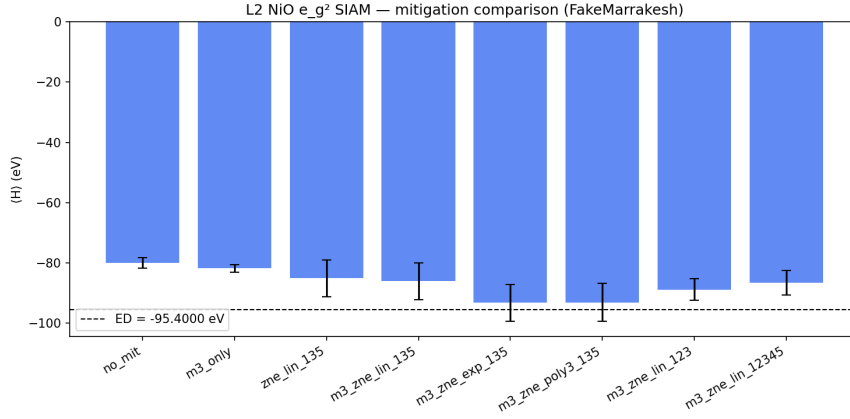


Figure 2: L2 mitigation comparison: per-config $\langle H \rangle \pm \sigma$ against the ED reference.

7. Phase L3 — full d-shell ground state via ADAPT-VQE

L3 is the full Ni 3d (5) + ligand bath (5) active space — 20 spin-orbitals, **18 qubits** after tapering — solved with ADAPT-VQE (99-operator pool, gradient threshold 10^{-4} eV, 4 seeds).

- **Result:** $E_{VQE} = 267.47700$ vs $E_{ED} = 267.47700 \rightarrow |\Delta E| = 9.1 \times 10^{-11}$ eV. ADAPT selected **7 operators**; wall time 1543 s; multistart spread 3.08 eV (soft-cluster Layer-4 pass — perturbed seeds break the ${}^3A_{2g}$ symmetry).
- **Observables (all match ED to 8+ decimals):** $\langle n_d \rangle = 8.002$, $\langle n_p \rangle = 9.998$; per-orbital $n_{d_{3z^2}} = 1.001$, $n_{d_{x^2-y^2}} = 1.001$, $n_{d_{xz}} = n_{d_{yz}} = 2.000$ — the textbook ${}^3A_{2g}$ d^8 configuration (full t_{2g} , half-filled e_g).

The foundation passed an atomic-limit gate first: the impurity-only ED reproduces the closed-form Racah d^8 multiplet spectrum (${}^3F, {}^1D, {}^3P, {}^1G, {}^1S$) to < 1 meV with the correct [7, 5, 3, 9, 1] degeneracies. (Eight plan-level physics bugs were caught during implementation by the two-stage review loop.)

8. Phase 5 — qEOM excited states and L-edge XAS

On the L3 ground state, a core hole is introduced ($H' = H + V_{\text{core}}$, $V_{\text{core}} = -U_{dc} \sum n_d$, $U_{dc} = 8.5$ eV); the core-excited state ψ'_{GS} ($d^9 {}^2E_g$) is found by ADAPT on H' , and **Bauer qEOM** generates

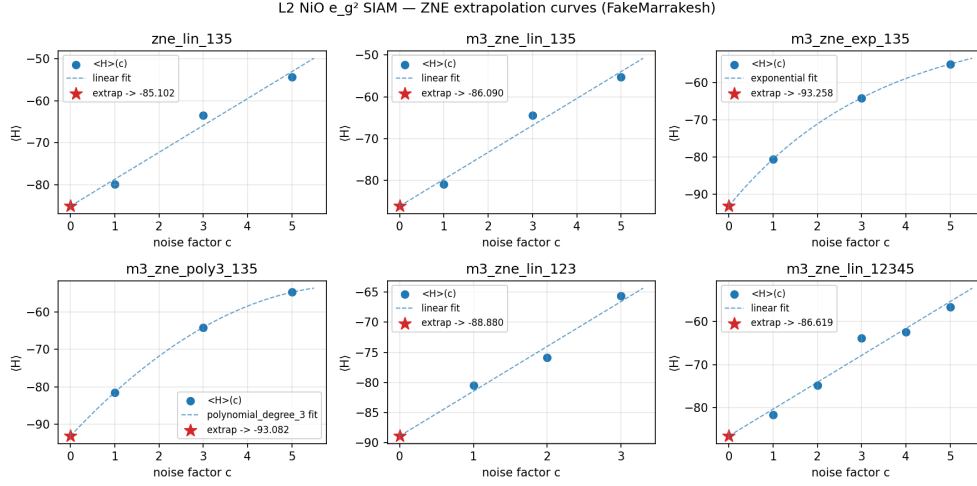


Figure 3: L2 ZNE extrapolation curves: $\langle H \rangle(c)$ vs noise factor with fits extrapolated to $c = 0$.

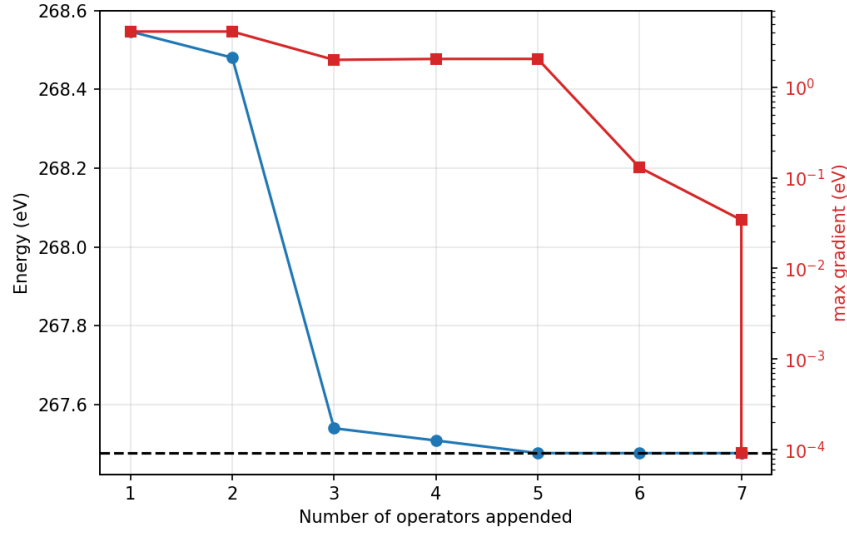


Figure 4: L3 ADAPT-VQE convergence trace: energy and selected-operator gradient vs iteration.

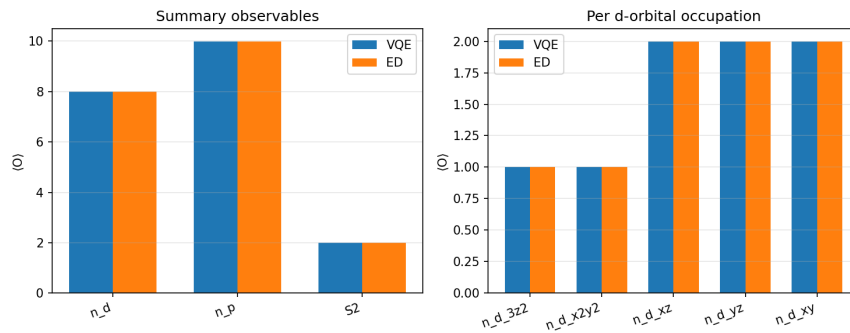


Figure 5: L3 ground-state observables: VQE vs ED occupations.

the excitation manifold. The L-edge XAS cross-section is assembled from cross-sector dipole matrix elements:

$$\sigma_q(\omega) = \sum_F |\langle F | D_q | \psi_{GS} \rangle|^2 \mathcal{L}(\omega; \Gamma = 0.5 \text{ eV}), \quad q \in \{\text{lin_z}, \text{lin_xy}\}.$$

- $E_{GS} = 267.477$, $E'_{GS} = 256.864$ eV; leading peak at $E'_{GS} - E_{GS} = -10.613$ eV (shifted to $\omega_0 = 0$).
- All five validation gates pass on **both** polarisation channels; L_2 distance vs the Path-B ED reference $\approx 4 \times 10^{-6}$. Leading-peak weights: `lin_z` 0.999, `lin_xy` 0.749.
- A qEOM technicality (Observation #20): an anti-Hermitian (UCC-style) pool makes the commutator-metric overlap $S \equiv 0$; the qEOM uses a non-antisymmetric hopping pool, decoupled from the ADAPT optimisation pool.
- **Open:** the independent Path-A EDRIXS cross-check (Docker) confirms single-peak structure but shows a $\sim 40\%$ spectral-weight gap vs Path B (orbital-channel vs photon-polarisation decompositions, spin-resolution, thermal averaging) — deferred to v2.

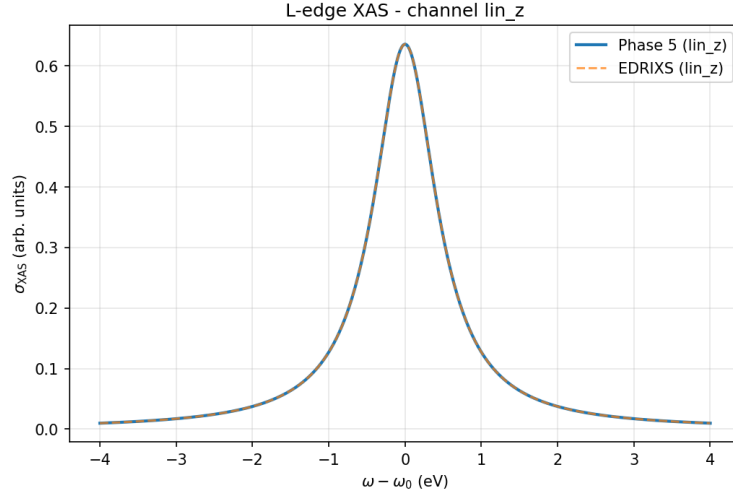


Figure 6: L-edge XAS, `lin_z` channel.

9. Phase 5b — hardware execution track

The goal of Phase 5b was to run the manual M3 + digital-ZNE spectroscopy stack on real IBM hardware.

Task 3e — execution-layer migration. `backend.run()` is removed in `qiskit_ibm_runtime` 0.44.1; the entire hardware-reachable path was unified on a single `_sample_counts(mode, ...)` seam over `qiskit_ibm_runtime.SamplerV2` (`mode` in `{BackendV2, Batch, Session}`). A real-hardware smoke confirmed dispatch.

L1 mitigated sweep — ISA-correctness. `MitigatedEstimator` was made ISA-correct for an opaque `Batch` (Approach A — *ISA-stable folding*): inputs are ISA-transpiled inside `run()` with a **pinned layout**, ZNE folding happens at the ISA level (odd integer factors, barriers, explicit-basis 1-qubit re-translation, connectivity- asserted), and the hardware path never re-routes — so M3 calibration stays valid across all per-term Hadamard circuits. A `--configs` driver runs the 4-way comparison; each config now runs in its **own** `Batch` with per-config error isolation.

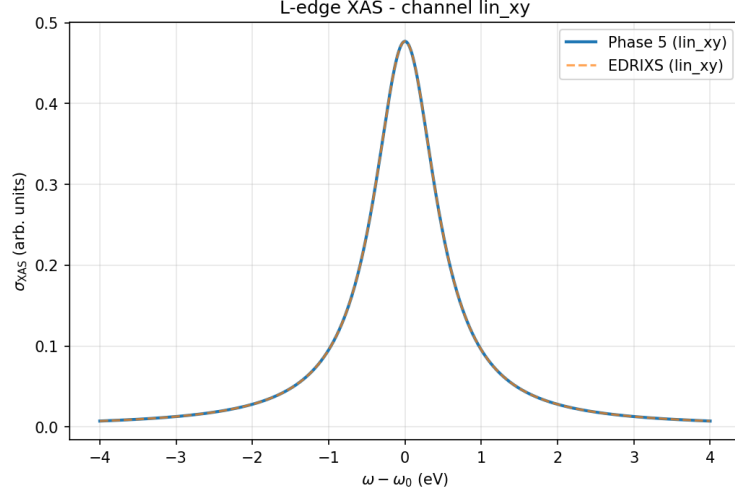


Figure 7: L-edge XAS, lin_xy channel.

Real ibm_marrakesh result. All 4/4 configs **ISA-dispatched** with no transpile/Batch errors — the spec’s hard success gate (dispatch-correctness) is met. The **physics is a clean negative result**, confirmed across two runs: the L1 cross-sector matrix element $|\langle \psi'_{GS} | D | \psi_{GS} \rangle|^2$ collapses from the exact 0.894 to ~ 0.001 – 0.005 across *every* config, and M3 / ZNE / M3+ZNE do **not** recover it. Even un-mitigated is ≈ 0 , so this is decoherence at this circuit depth, not a mitigation bug; the “effectiveness pass” is degenerate (all ratios ≈ 1.0 , ordering flips run-to-run $\rightarrow$ noise). Per the spec, dispatch is gated, agreement is reported.

config	run 1 $ \text{amp} ^2$	run 2 $ \text{amp} ^2$
no_mit	0.00527	0.00108
m3_only	0.00087	0.00039
zne_lin_135	0.00064	0.00149
m3_zne_poly3_135	(session dropped)	0.00194
exact	0.89380	0.89380

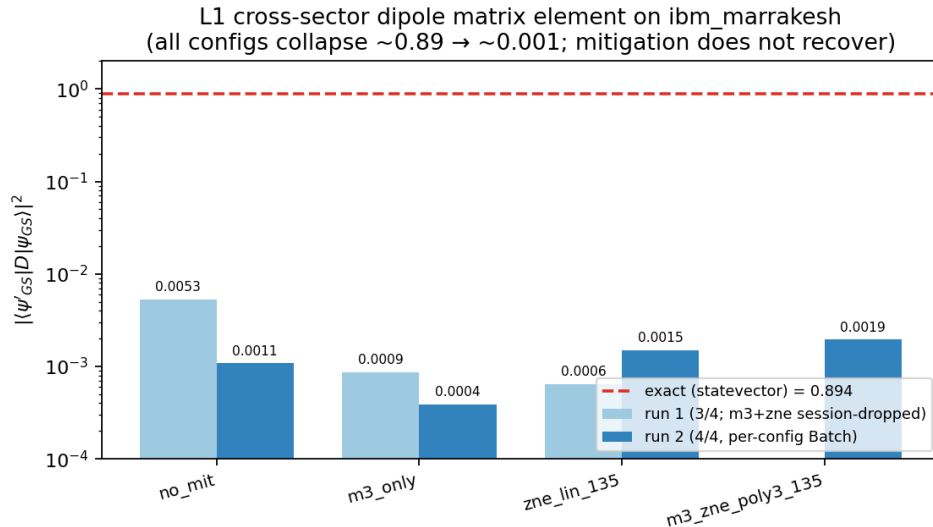


Figure 8: L1 hardware mitigated sweep on ibm_marrakesh: both runs vs the exact statevector value (log scale). The matrix element is decoherence-limited; mitigation does not recover it.

- **Tests.** Full suite **256 passed / 1 xfailed**; `ruff + mypy` gated in CI (package path). Every phase was implemented test-first via subagent-driven development with two-stage (spec + code-quality) review.
- **Process lessons captured this cycle** (observation log, for the team’s research-engineering discipline):
 - Gate shared-module changes on the **full** suite, not just the edited module’s tests (a regression in a *consumer* test module was missed by per-module runs and caught only by the full-suite gate).
 - Read a “must input be in form Y?” discriminator off the component that *enforces* Y (the execution `mode`/`Batch`), not a look-alike that merely carries the spec (a transpile target’s `coupling_map`).
 - Single-instance test fixtures hide cross-instance bugs (a 1-term observable masked a multi-term M3 calibration-layout bug); match fixture multiplicity to where state is shared.
 - Isolate each unit of a long remote/hardware sweep in its own session + capture per-unit errors; one shared session won’t survive the whole run.

11. Roadmap

- **Qiskit 2.0 migration** — the real fix for the remaining dependency-internal deprecations (Pulse package; the `stevedore/calc_final_ops` plugin-load log noise on hardware runs). Currently silenced via scoped `filterwarnings`.
- **Phase 6 — qEOM RIXS map** $I(\omega_{\text{in}}, \omega_{\text{loss}})$ via the Kramers–Heisenberg cross-section; the Phase-5 primitives (core hole, dipole, qEOM, cross-sector matrix element, Lorentzian assembly) are the inputs.
- **Path-A EDRIXS reconciliation** — resolve the ~40% spectral-weight gap (spin / thermal / convention).
- **L2 hardware demo** (Phase 5c).
- **Real signal on hardware** — the L1 observable is decoherence-limited; a future attempt needs a shallower state-prep and/or dynamical decoupling.

Appendix — references

- Haverkort, Zwierzycki, Andersen, *Phys. Rev. B* **85**, 165113 (2012).
- EDRIXS pedagogical examples 3 and 6.
- Repository: `liamlts/research-cowork`, package `siam_vqe`; main @ 648c125c.
- Phase PRs: #1 (L2 noise study), #4 (Phase 5b hardware), #5 (`_retranslate_to_basis` Qiskit-2.0 cleanup). Per-phase specs, plans, and closeout notes under `siam_vqe/docs/superpowers/`.